2.4 Graphs

<table>
<tr><th>Subject content</th><th></th><th>Guidance</th><th>Curriculum reference</th></tr>
<tr><td>A</td><td>Recognise, name and plot graphs parallel to the axes and the graphs of $y = x$ and $y = -x$</td><td>Plot the graphs:
$x = 4$
$y = -3$</td><td>A7.3A</td></tr>
<tr><td>B</td><td>Plot linear graphs using a table of values</td><td>Plot the graph $y = 2x + 3$<table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td></td><td></td><td>7</td><td></td><td></td></tr></table></td><td>A7.3B</td></tr>
<tr><td>C</td><td>Find the midpoint of a line segment, given two coordinates</td><td>Find the midpoint of a line with ends at $(-2, 3)$ and $(6, 10)$</td><td>A7.3D</td></tr>
<tr><td>D</td><td>Solve problems involving coordinates and straight lines</td><td>Given three coordinates A, B, C, write down the coordinates of D such that $ABCD$ is a parallelogram.</td><td>A7.3E</td></tr>
<tr><td>E</td><td>Recognise, draw and interpret graphs showing constant rates of change</td><td>A tap runs at a constant rate. After 8 minutes, the tap has run 75 litres of water.
Plot this information on a graph.
Use your graph to estimate how long it took to run 40 litres of water.</td><td>A7.3C
A8.3A
A9.5A</td></tr>
<tr><td>F</td><td>Draw, use and/or interpret distance-time graphs</td><td>Vladimir leaves home at 08 00
He travels for 30 minutes at a constant speed until he reaches a shop, 3 km away.
He stays at the shop for 45 minutes, then returns home arriving at 09 35
Plot this information on a distance-time graph.
Which part of his journey was the fastest?
Explain how you know this from your graph.</td><td>A8.3B</td></tr>
<tr><td>G</td><td>Solve problems by drawing and interpret linear conversion graphs</td><td>Currency conversion graphs</td><td>A8.3A
A8.3C</td></tr>
<tr><td>H</td><td>Solve problems by drawing and interpreting real-life non-linear graphs</td><td>Read values and describe trends from a real-life graph</td><td>A7.3C
A8.3D</td></tr>
</table>

2.4 Graphs *continued*

Subject content		Guidance	Curriculum reference
I	Solve problems in context by drawing and interpreting real-life linear and quadratic graphs	Use the distance-time graph to work out Vladimir's speed on his journey home.	A8.3D A8.3H A9.5G
J	Calculate the gradient and y-intercept of a linear graph of the form $y = mx + c$	Find the gradient and the y-intercept of the graph of: $y = 3x - 2$	A8.3F A9.5B
K	Find the equation of a straight line graph in the form $y = mx + c$		A8.3G
L	Recognise and draw and/or sketch graphs with equations of the form $y = mx + c$ and $ax + by = c$	Draw the graphs: $y = 3x - 2$ $5x + 6y = 30$	A8.3G A9.5B A9.5D
M	Compare linear graphs using gradients and y-intercept	Write down the equation of a line that is parallel to $y = 3x + 2$	A9.5B A9.5C
N	Plot graphs of quadratic functions using a table	Plot the graph $y = 2x^2 - 3$	A9.5F
O	Solve a pair of linear simultaneous equations by drawing graphs	Plot the graphs: $y = 3x + 2$ $y = 5x + 6$ and hence find a solution to the simultaneous equations: $y = 3x + 2$ $y = 5x + 6$	A9.5E